

1. กำหนด Node และ Relationship

ใช้โครงสร้างดังนี้

- **Node:** Student, Club, Interest
- **Relationship:** FRIEND_OF (Student $\leftrightarrow$ Student), MEMBER_OF (Student $\rightarrow$ Club), INTERESTED_IN (Student $\rightarrow$ Interest), FOCUSES_ON (Club $\rightarrow$ Interest)
- **เป้าหมาย:** แนะนำชมรมโดยดูทั้งจำนวนเพื่อนที่เป็นสมาชิก และดูว่าชมรมนั้นตรงกับความสนใจของนักเรียนหรือไม่

Pattern หลักสำหรับนับเพื่อน:

```
(me:Student)-[:FRIEND_OF]-(friend:Student)-[:MEMBER_OF]->(club:Club)
```

คำอธิบาย: เดินจากนักเรียนไปหาเพื่อน แล้วดูว่าเพื่อนแต่ละคนอยู่ชมรมใด ส่วน Interest จะนำมาใช้เสริมเหตุผลการแนะนำ

รายละเอียด Entity และ Attribute

- **Student (Node):**
 - Properties: student_id (Primary Key), name
- **Club (Node):**
 - Properties: club_id (Primary Key), name
- **Interest (Node):**
 - Properties: name

Relational Schema:

- **Student** (student_id, name)
- **Club** (club_id, name)
- **Interest** (name)
- **Friendship** (student_id_1, student_id_2)

- Membership (student_id, club_id, joined_date, role)
- Student_Interest (student_id, interest_name)
- Club_Interest (club_id, interest_name)

2. สร้างข้อมูลตัวอย่าง (นักเรียน 6 คน ชมรม 4 ชมรม)

```
CREATE
(s1:Student {student_id: 'S001', name: 'Earth'},
(s2:Student {student_id: 'S002', name: 'Yamal'},
(s3:Student {student_id: 'S003', name: 'View'},
(s4:Student {student_id: 'S004', name: 'Messi'},
(s5:Student {student_id: 'S005', name: 'Kade'},
(s6:Student {student_id: 'S006', name: 'Tar'},

(c1:Club {club_id: 'C001', name: 'Music Club'},
(c2:Club {club_id: 'C002', name: 'Painting Club'},
(c3:Club {club_id: 'C003', name: 'Basketball Club'},
(c4:Club {club_id: 'C004', name: 'Soccer Club'},

(i1:Interest {name: 'Strategy'},
(i2:Interest {name: 'Art'},

(s1)-[:FRIEND_OF]->(s2),
(s1)-[:FRIEND_OF]->(s3),
(s1)-[:FRIEND_OF]->(s4),
(s1)-[:FRIEND_OF]->(s5),

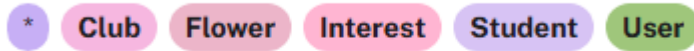
(s2)-[:MEMBER_OF]->(c1),
(s3)-[:MEMBER_OF]->(c1),
(s4)-[:MEMBER_OF]->(c1),
(s5)-[:MEMBER_OF]->(c2),
(s6)-[:MEMBER_OF]->(c3),
(s1)-[:MEMBER_OF]->(c3),

(s1)-[:INTERESTED_IN]->(i1),
(c1)-[:FOCUSES_ON]->(i1),
(c2)-[:FOCUSES_ON]->(i2);
```

คำอธิบาย: สร้างข้อมูลนักเรียน 5 คน (Earth, Yamal, View, Messi, Kade) และชมรม 3 ชมรม (Music, Painting, Basketball) โดยกำหนดความสัมพันธ์ :MEMBER_OF พร้อมเก็บบันทึก joined_date และ role ไว้บนตัว Relationship

Database information

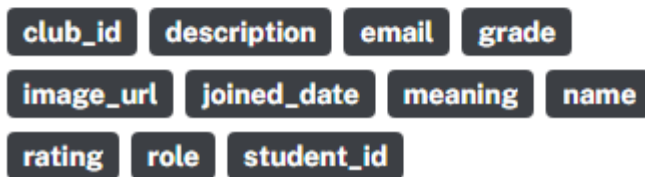
Nodes (12)



Relationships (13)



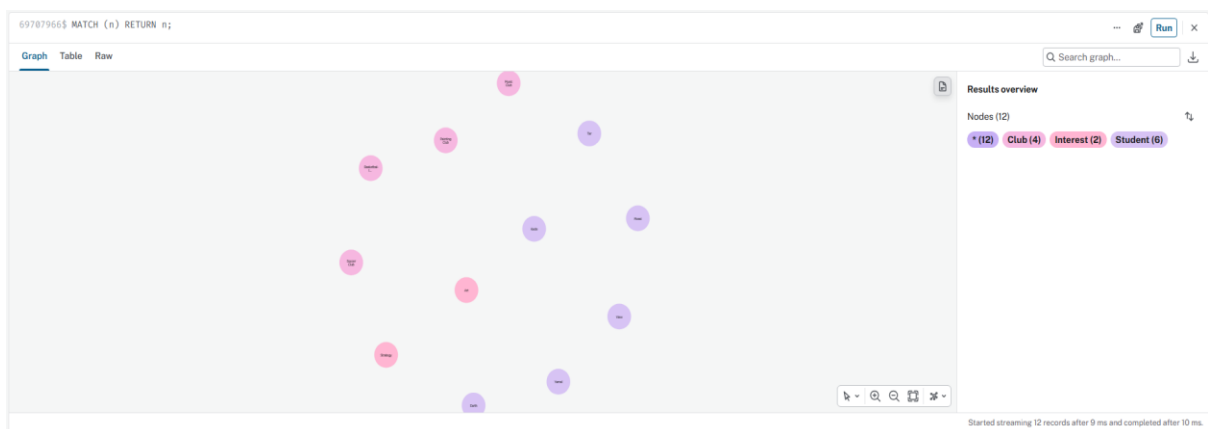
Property keys



3. ตรวจสอบข้อมูลทั้งหมด

MATCH (n)

RETURN n;



4. ดูชมรมของเพื่อนนักเรียน S001

```
MATCH (me:Student {student_id: 'S001'})-[:FRIEND_OF]-(friend:Student)-[:MEMBER_OF]->(club:Club)
RETURN friend.student_id AS Friend_ID,
       friend.name AS Friend_Name,
       club.club_id AS Club_ID,
       club.name AS Club_Name;
```

69707966\$ MATCH (me:Student {student_id: 'S001'})-[:FRIEND_OF]-(friend:Student)-[:MEMBER_OF]->(club:Club) RETURN friend.student_id AS Friend_ID, friend.name AS Friend_Name, club.club_id AS Club_ID, club.name ...  

TableRaw

Search table...

Graph chip



Friend_ID	Friend_Name	Club_ID	Club_Name
1 "S002"	"Yamal"	"C001"	"Music Club"
2 "S003"	"View"	"C001"	"Music Club"
3 "S004"	"Messi"	"C001"	"Music Club"
4 "S005"	"Kade"	"C002"	"Painting Club"

Started streaming 4 records after 33 ms and completed after 34 ms.

5. นับคะแนนเพื่อน

```
MATCH (me:Student {student_id: 'S001'})-[:FRIEND_OF]-(friend:Student)-[:MEMBER_OF]->(club:Club)
RETURN club.club_id AS Club_ID,
       club.name AS Club_Name,
       count(DISTINCT friend) AS Friend_Score
ORDER BY Friend_Score DESC;
```

69707966\$ MATCH (me:Student {student_id: 'S001'})-[:FRIEND_OF]-(friend:Student)-[:MEMBER_OF]->(club:Club) RETURN club.club_id AS Club_ID, club.name AS Club_Name, count(DISTINCT friend) AS Friend_Score ORDER B ...

Run

✕

TableRaw

Search table...

Graph chip

Download

Share

Club_ID	Club_Name	Friend_Score
1 "C001"	"Music Club"	3
2 "C002"	"Painting Club"	1

Started streaming 2 records after 64 ms and completed after 66 ms.

6. ดัดชมรมที่ตัวเองเป็นสมาชิกอยู่แล้ว

```
MATCH (me:Student {student_id: 'S001'})-[:FRIEND_OF]-(friend:Student)-[:MEMBER_OF]->(club:Club)
WHERE NOT EXISTS {
  MATCH (me)-[:MEMBER_OF]->(club)
}
RETURN club.club_id AS Club_ID,
       club.name AS Recommended_Club,
       count(DISTINCT friend) AS Friend_Score
ORDER BY Friend_Score DESC;
```

69707966\$ MATCH (me:Student {student_id: 'S001'})-[:FRIEND_OF]-(friend:Student)-[:MEMBER_OF]->(club:Club) WHERE NOT EXISTS { MATCH (me)-[:MEMBER_OF]->(club) } RETURN club.club_id AS Club_ID, club.name AS Re ...

Run

TableRaw

Search table...

Graph chip

Club_ID	Recommended_Club	Friend_Score
1 "C001"	"Music Club"	3
2 "C002"	"Painting Club"	1

Started streaming 2 records after 72 ms and completed after 74 ms.

7. เช็คความสนใจ Interest

```
MATCH (me:Student {student_id: 'S001'})-[:INTERESTED_IN]->(i:Interest)-[:FOCUSES_ON]-(club:Club)
RETURN club.name AS Club_Name,
       i.name AS Shared_Interest;
```

69707966\$ MATCH (me:Student {student_id: 'S001'})-[:INTERESTED_IN]->(i:Interest)-[:FOCUSES_ON]-(club:Club) RETURN club.name AS Club_Name, i.name AS Shared_Interest; ...  

Table	Raw
	Search table...  Graph chip  
Club_Name	Shared_Interest
1 "Music Club"	"Strategy"

Started streaming 1 record after 37 ms and completed after 38 ms.